Software Requirements Specification

Project: Study-path advisor

Group: (7) Musaed Al-Fareh, Melat Assefa, Alexia Cobzaru, Raluca Costache, Mohamed Elshami, Bilgin Eren

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1. Introduction

This Software Requirements Specification (SRS) describes the Study-Path Advisor, an intelligent system that helps university students turn their interests into concrete study paths. It defines what the system must do and the conditions under which it must operate, building on the problem and objectives laid out in Task 1 and the data and models developed in Task 2. Section 2 highlights the user requirements. Section 3 translates them into functional and non-functional system requirements. Section 4 identifies the requirements that exist because the system contains an AI component.

1.1. Purpose

Students choose electives, extracurricular activities and self-study largely on their own. Hence, many only realise near the end of their studies that their choices didn't prepare them adequately for the job or master's programme they plan on pursuing. The information needed to choose well exists but it is quite scattered and inconsistent. A student's interest, a course description and an official skill may not always use the same words so no fixed rule may connect them. The Study-Path Advisor seeks to address this specific issue by bridging this gap. When a student describes an interest, the system returns a few possible study paths, each with concrete courses, the skills that are gained from those courses and the occupation those skills lead to. The system helps to narrow and explain options but doesn't make decisions on the student's behalf. This SRS covers the complete system: the interface, the AI components (a course-category classifier and course-skill to ESCO matcher), the rule based steps that assemble paths and look up occupations, as well as the pipeline used to update data and models.

1.2. Intended audience

The developers in this Group
- Use it to design and build components

The client (Lecturer)
- Use it to confirm that the system's scope matches the problem agreed in Task 1

Students
- The system's primary users can read Section 2 to check that the user requirements reflect what they need when planning their studies

Programme coordinators
- As secondary users, can assess what the tool will provide for students (or won't)

1.3. Additional information

Data Sources:
- Courses are sourced from a scraped online-course catalogue covering Coursera, FutureLearn, Udacity and Simplilearn

Scope constraint:
- The Tilburg University course catalogue couldn't be obtained. University electives, etc. are therefore out of scope and the online catalogue stands in for the course layer

2. User requirements

Each user requirement states what a student can do with the system. "Traces to" links it to the Task 1 sub-problem (1.2.x) or objective (3.x) it serves. Parts marked Deferred are kept for traceability but cannot be met in this release, because the data they need (university regulations, master's programmes, previous students' choices) is not available; see 1.3.

ID	User requirement	Traces to
UR1	The system shall let a student describe an interest or desired destination in their own words, as free text.	1.2.4, 3.3
UR2	The system shall let a student enter their completed courses, preferred language, weekly study hours and the number of weeks available. *Deferred: programme, year, remaining credits and exchange preferences (need university regulations, see 1.3).*	1.2.4, 3.3

ID	User requirement	Traces to
UR3	The system shall give a student a shortlist of three to five study paths that match their stated interest and fit the context entered under UR2.	1.2.4, 3.3
UR4	The system shall show, for each suggested path, the courses it contains with their category, duration and the skills they teach. *Deferred: minors, projects, internships and exchanges (no data source).*	1.2.2
UR5	The system shall explain, for each course in a path, how it contributes to the student's stated interest or destination.	1.2.1, 3.4
UR6	The system shall state for each path whether it fits the student's context, and give the reason when a course was left out.	3.3, 3.4
UR7	The system should let a student mark courses they want to keep in their path. *Deferred: selecting master's programmes, minors and exchanges (no data source).*	1.2.4, 1.2.5
UR8	The system shall show how the suggested paths change when the student changes their interest, context or kept courses.	1.2.5, 3.3
UR9	The system shall show which occupations each path leads to. *Deferred: master's programmes and their entry requirements (no data source).*	1.2.1, 3.3
UR10	The system shall show which skills each path develops and which skills needed for the student's destination it does not cover.	1.2.1, 3.3
UR11	The system shall show the duration of each course and of each path. *Deferred: prerequisites, deadlines and decision moments (need university regulations, see 1.3).*	1.2.3, 3.3
UR12	*Deferred: the system could use previous students' choices to give context on possible paths. No such data exists yet; it would come from the system's own use logs.*	3.2

3. System requirements

3.1. Functional requirements

What the system does for the student, independent of how it is built. Priority uses MoSCoW; "Traces to" links each function to the user requirements and the Task 1 sub-problem or objective it serves. Thresholds are set in Section 4, which also identifies the functions that depend on the AI component.

ID	Feature / Function	Description	Priority	Action / Result	Traces to
FR-1	Capture stated interest	Accept an interest or destination as free text in the student's own words.	Must	Accepts 1–500 characters of free text and keeps it for the session. Empty input is refused with "Please describe an interest".	UR1; 1.2.4
FR-2	Capture study context	Record preferred language, weekly study hours, weeks available and completed courses.	Must	Stores language, hours per week (1–40), weeks available (1–52) and completed courses for the session. A missing field is shown as "not given" and its check in FR-5 is skipped.	UR2; 3.3
FR-3	Derive required skills	If the interest names an occupation, match it to ESCO and take the skills linked to it.	Must	Returns up to 3 ESCO occupations above the threshold and the union of their essential skills. If none clears it, shows "no occupation matched" and FR-4 uses the interest text alone.	UR3, UR10; 1.2.1
FR-4	Retrieve courses	Find catalogue courses whose skills (FR-15) match the required skills or the stated interest, narrowed by category (FR-14).	Must	Returns up to 50 candidate courses, each with title, category (FR-14) and matched ESCO skills (FR-15). If none are found, shows "no matching courses" and asks the student to rephrase.	UR3, UR4; 1.2.2
FR-5	Assess feasibility	Drop completed courses and courses in a non-preferred language or above the weekly hours; missing values stay, labelled "not listed".	Must	Removes completed courses, courses in another language and courses above the weekly hours, listing each removal with its reason. Courses with a missing value are kept and labelled "not listed".	UR6; 3.4
FR-6	Compose whole paths	Combine remaining courses with a listed duration into paths that, taken one after another, fit within the weeks available.	Must	Each path has at least one course and a total duration within the weeks available. Courses without a listed duration are left out of paths, and the student is told how many.	UR3; 1.2.2
FR-7	Rank and return paths	Rank by fit: share of required skills covered (FR-3), or interest-to-course similarity if no occupation matched.	Must	Returns 3–5 paths ordered by fit score (0–1), highest first; ties go to the shorter path. If fewer than 3 fit, returns those with "only N paths fit your constraints".	UR3; 3.3
FR-8	State what a path opens	List the occupations each path's ESCO skills lead to.	Must	Lists per path up to 5 ESCO occupations whose essential skills it covers most, with the share covered; otherwise "none matched".	UR9; 1.2.1
FR-9	State what a path closes off	List the required skills (FR-3) each path does not cover.	Should	Lists per path the required ESCO skills (FR-3) it does not cover. Shown only when an occupation was matched.	UR10; 1.2.1
FR-10	Justify each path	Show which skills each course contributes and	Should	Lists per course the ESCO skills it contributes with their similarity scores, and per path the fit score used in FR-7.	UR5; 3.4

ID	Feature / Function	Description	Priority	Action / Result	Traces to
		each path's fit score.			
FR-11	Show course duration	Show the duration of each course and the total duration of each path.	Should	Shows each course's listed duration and each path's total in weeks; a missing duration is shown as "not listed".	UR4, UR11; 1.2.3
FR-12	Re-evaluate on change	When the student edits the interest or context, recompute FR-3 to FR-11.	Should	Recomputes the shortlist when the interest, context or kept courses change; unchanged inputs are kept and not re-entered.	UR7, UR8; 1.2.5
FR-13	Withhold low-confidence matches	Treat any category confidence (FR-14) or similarity score (FR-3, FR-4, FR-15) below its threshold, set in Section 4, as no match.	Must	Below threshold, a category is shown as "uncategorised" and a skill or occupation as "unmatched". Such items are never shown as matches and are not used in ranking.	UR5, UR10; 3.4
FR-14	Classify courses	Assign each uncategorised course to one of the 11 categories, with a confidence score.	Must	Each course gets one of the 11 categories and a confidence (0–1), or "uncategorised" below threshold. Runs in batch when the catalogue is loaded.	UR4; 3.4
FR-15	Map course skills to ESCO	Match each free-text course skill to its closest ESCO skills, with similarity scores.	Must	Each course skill gets up to 5 ESCO skills with a similarity score (0–1). Matches below threshold are dropped; a skill with none left is marked "unmatched".	UR10; 3.4
FR-16	Update data and models	Let a maintainer load new data, retrain the classifier and rebuild the ESCO skill index.	Could	New data and models replace the old ones only after passing the acceptance criteria in Section 4; otherwise the previous version stays in use.	3.4
FR-17	Report a wrong match	Let the student flag a skill or occupation shown for a course or path as wrong.	Could	Stores the flagged item, its course or path and a timestamp, with no personal data (NFR-8). The student sees "flagged for review".	UR10; 3.4

3.2. Non-functional requirements

Non-functional requirements specify the quality attributes, operational limits, external constraints, and ethical boundaries under which the system must operate. They govern how the functional requirements are implemented and delivered.

ID	Category	Requirement Description	Success Metric/ Constraint	Traces to
NFR-1	Performance and Latency	The system must return recommendations and recalculate paths within an acceptable interactive timeframe to preserve a smooth user interface.	Shortlist generation and dynamic re-evaluation must return results within <3 seconds after query submission	FR-1 to FR7, FR12
NFR-2	Availability and Scalability	The core path-generation service must remain available to students during study planning and registration peaks.	Minimum 99% uptime during academic term decision periods. Supports concurrent student recommendation queries without request drops	FR-1, FR-2, FR-7, FR-12
NFR-3	Data Completeness and Graceful Degradation	Because scraped third-party online catalogues contain irregular fields (e.g. missing course duration), the system must handle null values predictably without crashing.	Missing fields must be explicitly labelled as "unknown"/ "not listed"	FR-2, FR-5, FR-6, FR-11
NFR-4	Classifier and Matcher Inference Efficiency	Model inference for category assignment, ESCO occupation linking, and vector similarity lookup must run efficiently within the recommendation loop.	Offline batch pre-processing for static catalogue courses (FR-14, FR-15) and real-time occupation and similarity lookups must resolve in <500ms per query	FR-3, FR-4, FR-15
NFR-5	Maintainability and Extensibility	The system pipeline must allow catalog updates, model re-training and ESCO indexing without modifying downstream path-generation logic.	Pipeline supports incremental index rebuilds and model substitution via standard modular APIs or CLI tools.	FR-14, FR-15, FR-16
NFR-6	Transparency and Explainability	The system must make recommendation rationales interpretable and avoid opaque decision-making. It must present itself as an advising aid rather than an authoritative automated decider.	Every returned path must display its specific ESCO skill links, skill gaps, fit score and matching confidence. Unsupported options must state the reason for exclusion.	FR-7, FR-8, FR-9, FR-10, FR-13
NFR-7	Ethical Safety and Bias Mitigation	Path generation must not introduce demographic, gender, or institutional bias into career or course recommendations, relying purely on academic background, skills and student specified interests.	Semantic embeddings and classification rules must be regularly audited against occupational stereotyping. Low-confidence matches must be suppressed rather than guessed	FR-3, FR-4, FR-13, FR-17
NFR-8	Privacy and Session Boundary	Personal study preferences, entered courses and session queries must be handled securely in compliance with GDPR.	Inputs held in memory for the duration of the student's active session only. No persistent profiling without explicit consent. Error reports (FR-17) must be anonymised.	FR-1, FR-2, FR-12, FR-17

ID	Category	Requirement Description	Success Metric/ Constraint	Traces to
NFR-9	External Dependency Management	The system relies on external taxonomy releases (ESCO) and heterogeneous scraped catalogues. Schema drift in external sources must not corrupt active recommendations	System must pin specific ESCO release versions and ingest scraped catalogue changes via an isolated staging and validation pipeline	FR-3, FR-4, FR-15, FR-16

4. AI specific requirements

The Study-Path Advisor uses AI to classify courses into the 11 defined categories and match free-text descriptions to ESCO skills and occupations. These components support course retrieval, path ranking and explanations. AI is needed because student interests, course descriptions and ESCO terminology may express similar concepts using different words.

ID	Requirement	Verification / acceptance criterion	Traces to
AIR-1	**Course classification.** The classifier will assign each course to one of the 11 defined categories and return a confidence score. If the score is below the category threshold, the course shall be marked “uncategorised”.	Each processed course has a category and confidence score, or an “uncategorised” status. Classification quality is evaluated using macro-F1 and a confusion matrix on held-out labelled courses. Minimum macro-F1: TBD.	FR-13, FR-14
AIR-2	**Semantic matching.** The matcher will associate occupation descriptions and free-text course skills with ESCO entries using semantic similarity. Matches below the applicable threshold shall be rejected.	Every accepted match includes an ESCO identifier and similarity score. Matching precision and recall are measured against manually checked examples. Minimum acceptable values: TBD.	FR-3, FR-4, FR-13, FR-15
AIR-3	**Threshold validation.** The system will use separately configurable thresholds for course classification, occupation matching, course retrieval and course-skill matching. Thresholds shall be selected using validation data before final testing.	Threshold values and their selection method are documented. Tests confirm that scores below each threshold produce no accepted match. Final thresholds: TBD.	FR-13; NFR-7
AIR-4	**Evaluation data quality.** Model evaluation will use manually checked reference labels and matches. Training, validation and test data shall remain separate, with duplicate course records kept within the same split.	The dataset documentation records label definitions, split membership and duplicate handling. Final evaluation uses test examples that were not used for training or threshold selection.	FR-14, FR-15, FR-16
AIR-5	**Traceable explanations.** Recommendations will distinguish source information from AI-inferred categories and matches. Each accepted AI match used in an explanation shall retain its source text, ESCO identifier and score.	Displayed skill links can be traced to the corresponding course information and matching output. Similarity scores are identified as similarity measures rather than probabilities of correctness.	FR-7, FR-10, FR-13; NFR-6
AIR-6	**Bias evaluation.** Before a model version is adopted, recommendations shall be tested for occupational stereotyping using paired inputs that differ only in demographic wording unrelated to study preferences or constraints.	The evaluation records changes in recommended courses, occupations and rankings. Unjustified differences are reviewed and resolved before adoption.	FR-3, FR-4, FR-7; NFR-7
AIR-7	**Reproducibility and model updates.** Each model release will record the classifier version, semantic model version, ESCO release, catalogue version, thresholds and evaluation results. Replacement models shall satisfy the same acceptance criteria before activation.	A saved evaluation can be reproduced using the recorded configuration. Updated models preserve the interfaces expected by course retrieval and path generation.	FR-16; NFR-5, NFR-9
AIR-8	**Inference performance.** Static course classification and course-skill matching shall run during catalogue preparation. Real-time AI lookups shall satisfy the inference limit specified in NFR-4.	Timing tests on the documented deployment environment show that real-time AI lookups complete within 500 ms per query.	FR-3, FR-4, FR-14, FR-15; NFR-1, NFR-4

4.1. Relationship to functional requirements

FR-3, FR-4, FR-14 and FR-15 directly use AI-generated classifications or semantic matches. FR-7 and FR-10 depend on these outputs when ranking and explaining paths. FR-13 controls whether uncertain AI outputs are accepted, while FR-16 supports model maintenance.

Feasibility checks and path composition remain rule-based. When an occupation is matched, skill-coverage ranking can also be calculated deterministically from the accepted ESCO matches. FR-12 indirectly depends on AI because changing an interest can trigger new semantic lookups.